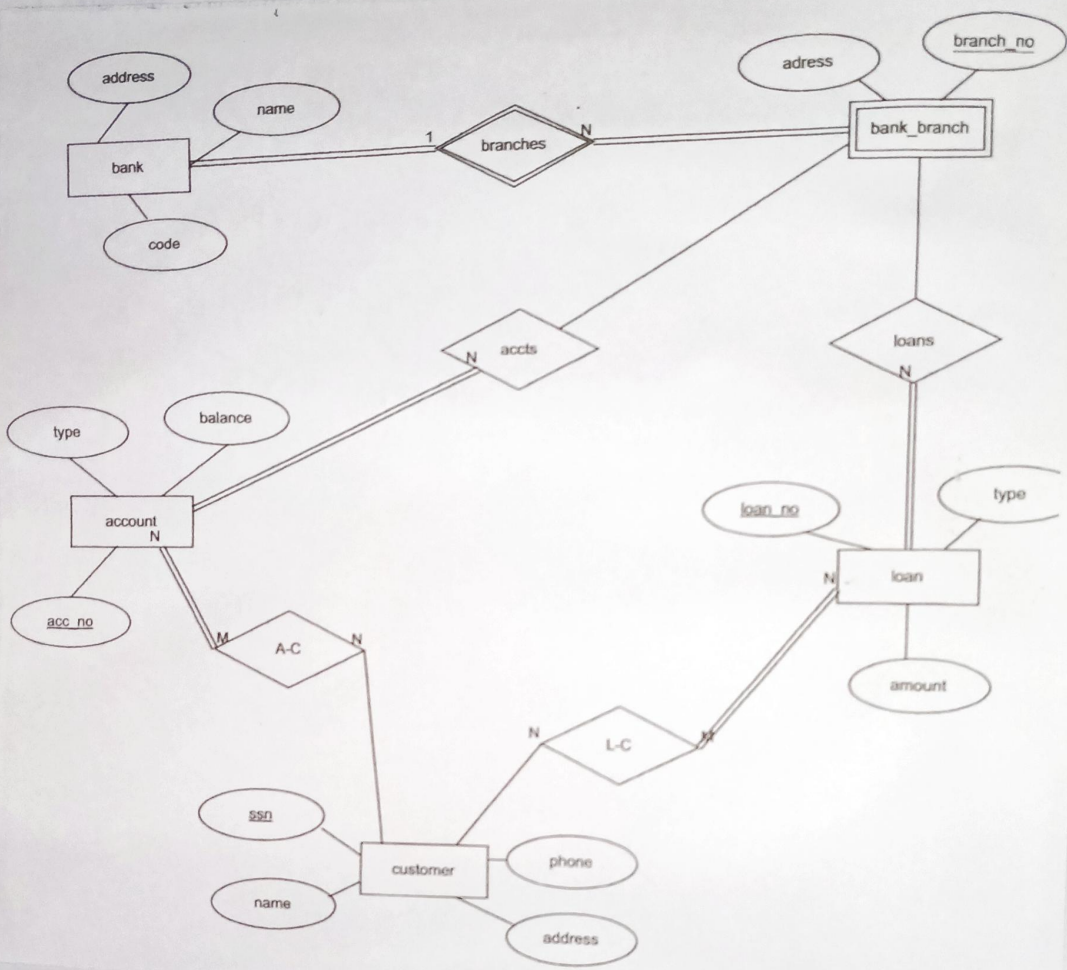






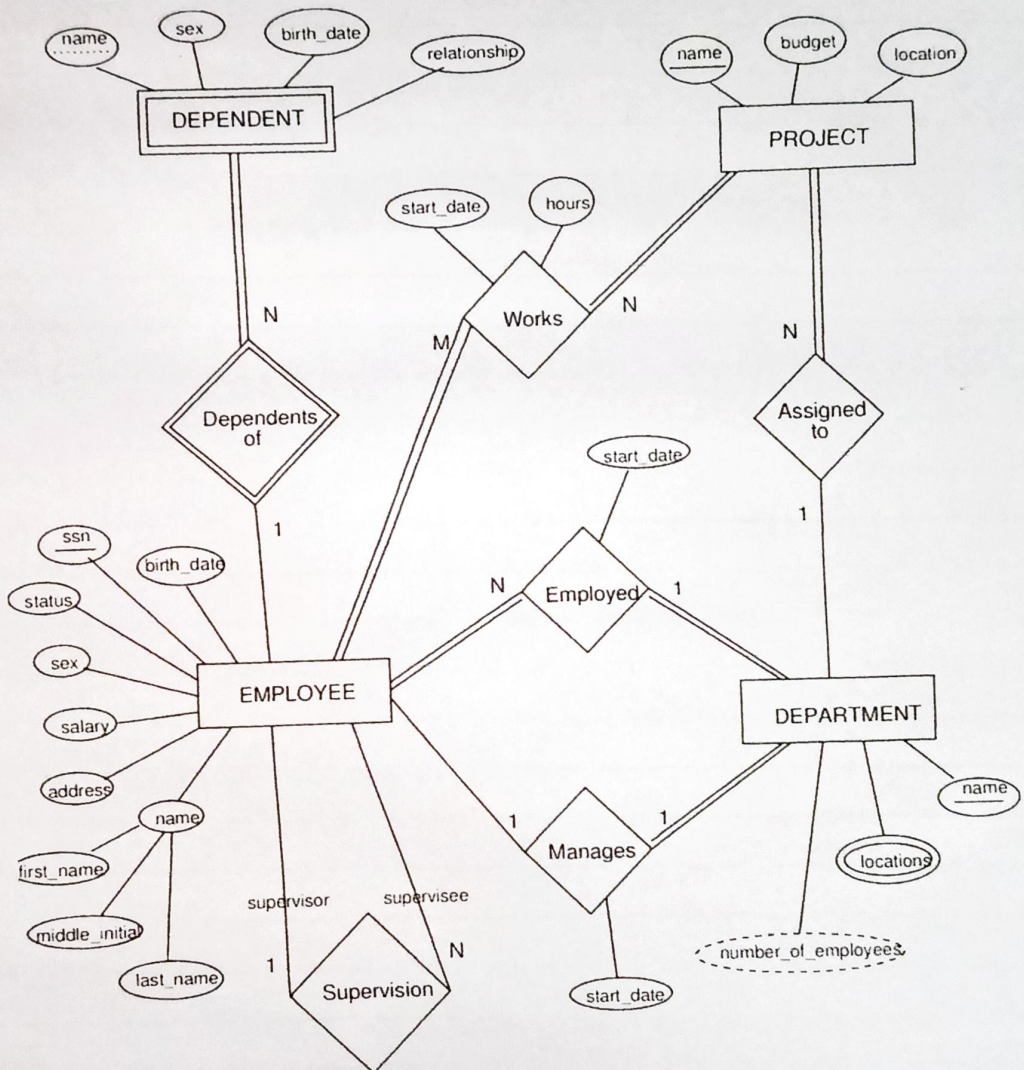
Date.....

Experiment No.....

18

Company

create ER diagram for Company database/bank using DrawIO.



Desc Employee;

Field	Type	Null	Key	Default	Extra
Fname	varchar(20)	NO		NULL	
Minit	varchar(1)	YES		NULL	
Lname	varchar(15)	NO		NULL	
Ssn	char(9)	YES	PRI	NULL	
Bdate	date	YES		NULL	
Address	varchar(50)	YES		NULL	
Sex	char(1)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	
Superssn	char(9)	YES		NULL	
Dno	int(11)	NO		NULL	

Desc department;

Field	Type	Null	Key	Default	Extra
Dname	varchar(25)	NO	UNI	NULL	
Dnumber	int(11)	NO	PRI	NULL	
Mgrssn	char(9)	YES		NULL	
Mgrstart_date	date	YES		NULL	



Date.....

Experiment No.....

1. Consider the Company database and create the below tables by properly specifying the primary keys and the foreign keys.

Employee (Fname: varchar, Minit: char, Lname: varchar, SSN: char, Bdate: Date, Address: varchar, Sex: char, Salary: decimal, Super-SSN: char, DNO: int).

Department (Dname: varchar, Dnumber: int, mgr-SSN: char, mgr-start-date: Date).

Dept-location (Dnumber: int, Dlocation: varchar)

Project (Pname: varchar, Pnumber: int, Plocation: varchar, dnum: int).

Works-on (ESSN: char, PNO: int, hours: decimal)

Dependent (ESSN: char, dependent-name: varchar, Sex: char, Bdate: Date, relationship: varchar).

Create Table Employee (

Fname varchar(20) Not null,

Minit varchar(1),

Lname varchar(15) Not null,

SSN char(9) Not null Primary Key,

Bdate date,

Address varchar(50),

Sex char(1),

Salary decimal(10,2),

Superssn char(9),

Dno int not null

);

Desc dependent;

Field	Type	NULL	Key	Default	Extra
Essn	char(9)	NO	PRI		
Department_name	varchar(18)	NO	PRI		
Sex	char(1)	YES		NULL	
Bdate	date	YES		NULL	
Relationship	varchar(8)	YES		NULL	

Desc dept-location;

Field	Type	NULL	Key	Default	Extra
Dnumber	Int	NO	PRI	NULL	
Dlocation	varchar(15)	NO	PRI	NULL	

Desc Project;

Field	Type	NULL	Key	Default	Extra
pname	varchar(25)	NO	UNI	NULL	
Pnumber	Int	NO	PRI	NULL	
plocation	varchar(15)	NO		NULL	
Dnumber	Int	NO		NULL	



```
Create table department (  
  Dname varchar(25) not null,  
  Dnumber int not null,  
  Mgrssn char(9),  
  mgrstart-date date,  
  Primary key (dnumber),  
  Unique (dname)  
);
```

```
Alter table department  
Add constraint foreign key (mgrssn) References  
Employee (ssn) on delete cascade on update cascade;
```

```
Create table dependent (  
  Essn char(9);  
  Dependent_name varchar(18),  
  Sex char,  
  Bdate date,  
  Relationship varchar(8),  
  Primary key (essn, dependent_name),  
  Foreign key (essn) References Employee (ssn)  
);
```

```
Create table dept_location (  
  Dnumber int,  
  Dlocation varchar(15),  
  Primary key (dnumber, dlocation),  
  Foreign key (dnumber) References department (dnumber)  
  on delete cascade on update cascade.  
);
```


Desc WORKS-ON;

Field	Type	NULL	Key	Default	extra
Essn	char(9)	NO	PRI	NULL	
pno	int	NO	PRI	NULL	
Hours	decimal(4,1)	NO		NULL	



```
Create table project (  
  Pname varchar(25) not null,  
  Pnumber int,  
  Plocation varchar(15),  
  Dnum int not null,  
  Primary Key (Pnumber),  
  Unique (Pname),  
  Foreign Key (dnum) references department(dnumber)  
);
```

```
create table works-on(  
  Essn char(9),  
  Pno int,  
  Hours decimal(4,1),  
  Primary Key (Essn, Pno),  
  Foreign Key (Essn) references employee (Ssn),  
  Foreign Key (Pno) references project (Pnumber)  
);
```


Select * from Employee;

Fname	Minid	Lname	Ssn	Bdate	Address	Sex	Salary	Superssn	Dno
Ankur	F	Singh	123451234	2004-08-12	Texas	M	70000.00	67812345	9
Manya	C	mya	12345555	1998-07-05	California	F	40000.00	123456789	1
John	B	Smith	12345678	1995-01-04	Houston	M	30000.00	45678321	5
Ritik	D	Rajput	45678321	1990-09-06	Houston	M	50000.00	123456789	6
Shravya	E	Sharma	67812345	2001-01-03	Houston	F	60000.00	12345555	8



2

Insert at least five tuple in each relation.

Insert into Employee values

('John', 'B', 'Smith', '12345678', '1995-01-04', 'Houston',
'M', 30000.00, '45678321', 5),
('Mariya', 'C', 'Moya', '12345555', '1998-07-05',
'31 California Houston', 'F', 40000.00, '12345678',
6),
('Ravi', 'D', 'Rajput', '45678321', '1990-09-06',
'70 Ottawa Houston', 'M', 50000.00, '123456789',
7),
('Shivanya', 'E', 'Sharma', '67812345', '2001-07-03',
'Stafford', 'F', 60000.00, '12345555', 8),
('Ankur', 'F', 'Singh', '12341234', '2004-08-12',
'Texas', 'M', 70000.00, '67812345', 9);

Insert into department (dname, dnumber, mgrssn,
mgrstart_date) values

('Research', 5, '12345678', '2023-01-01'),
('Finance', 6, '12345555', '2023-02-01'),
('IT', 7, '45678321', '2023-03-01'),
('Marketing', 8, '67812345', '2023-04-01'),
('Operations', 9, '12341234', '2023-05-01');

Select * from department;

Dname	Dnumber	Mgrssn	Mgrstart-date
Research	5	12345678	2028-01-01
Finance	6	12345655	2028-02-01
IT	7	45678821	2023-03-01
Marketing	8	67812345	2023-04-01
operations	9	12341234	2023-05-01

Select * from dependent;

Essn	dependent-name	Sex	Bdate	Relationship
12345555	Emma Moya	F	2018-07-15	child
12345678	Michael Smith	M	2015-02-10	child
12345678	Sarah Smith	mf	2016-10-20	Spouse
45678821	Aam Rajput	M	2020-03-30	child
67812345	Noah Sharma	M	2019-05-25	child.

Select * from dept-location;

Dnumber	Dlocation
5	Stafford
6	Los Angeles
7	chicago
8	Houston
9	San Francisco



Insert into dependent (essn, dependent_name, sex, bdate, relationship) values

('12345678', 'Sarah Smith', 'F', '2016-10-20', 'Spouse'),
('12345678', 'Michael Smith', 'M', '2016-02-10', 'child'),
('12345555', 'Emma Moya', 'F', '2018-07-15', 'child'),
('45678321', 'Liam Rajput', 'M', '2020-03-30', 'child'),
('67812345', 'Noah Sharma', 'M', '2019-05-25', 'child');

Insert into dept_location (dnumber, dlocation) values

(5, 'Stafford'),
(6, 'Los Angeles'),
(7, 'Chicago'),
(8, 'Houston'),
(9, 'San Francisco');

select * from Project;

Pname	Pnumber	Plocation	Dnum
Research project 1	1	Stafford	5
Research project 2	2	Stafford	5
IT Project 1	3	chicago	7
Marketing project 1	4	Houston	8
operation Project 1	5	San Francisco	9

select * from works-on;

Essn	Pno	hours
12341234	1	40.0
12345555	2	35.5
12345678	3	42.0
45678321	4	38.5
67812345	5	37.0



Date.....

Experiment No.....

Insert into project (pname, pnumber, plocation,
dnum) values

('Research Project 1', 1, 'Stafford', 5),

('Research Project 2', 2, 'Stafford', 5),

('IT Project 1', 3, 'chicago', 7),

('Marketing Project 1', 4, 'Houston', 8),

('Operations Project 1', 5, 'San Francisco', 9);

Insert into works-on (essn, pno, hours) values

('12345678', 1, 40.0),

('12345555', 2, 35.5),

('45678321', 3, 42.0),

('67812345', 4, 38.5),

('12345234', 5, 37.0);

1)

fname	lname	address
John	Smith	Houston

2)

pnumber	dnum	lname	address	bdate
1	5	Smith	Houston	1995-01-04
2	5	Smith	Houston	1995-01-04

3)

fname	lname	Sup-fname	Sup-lname
Ankur	Singh	Shivanya	Sharma
Mariya	Moya	John	Smith
John	Smith	Ritvik	Rajput
Ritvik	Rajput	NULL	NULL
Shivanya	Sharma	Mariya	Moya



3.

QUERIES.

1) Retrieve all employees who work for the 'Research' department.

Select e.fname, e.lname, e.address

From employee e

Join department d on d.dnumber = e.dno

where d.dname = 'Research';

2) For every project located in 'Stafford', list the project number, the controlling department number, and the department manager's last name, address, and birth date.

Select P.pnumber, P.dnum, e.lname, e.address, e.bdate

From project P, department d, employee e

where P.plocation = 'Stafford' And P.dnum = d.dnum

-ber and d.mgrssn = e.ssn;

3) For each employee, retrieve the employee's first & last name and the first & last name of his or her immediate supervisor.

Select e.fname, e.lastname, s.fname as sup_fname,
s.lname as sup_lname

From employee e

Left join employee s on e.superssn = s.ssn;

pnumber
1

fname	Minid	lname
Ankur	F	Singh
Mariya	C	Moya
John	B	Smith
Ritvik.	D	Rajput

fname	minit	lname	ssn	bdate	address
John	B	Smith	12345678	1995-01-04	Houston
		alno	Superssn	Salary	Sex
		5	45678321	30000.00	M

4. Make a list of all project numbers for projects that involve an employee whose last name is 'Smith'.

Select distinct, pnumber

From project P

Join works-on w on $p.number = w.pno$

Join employee e on $w.essn = e.ssn$

where $e.lname = 'Smith'$;

5. Retrieve all employees whose address is in Houston, Texas.

Select fname, minit, lname

From Employee

where address like '% Houston %' or address like '% Texas %' ;

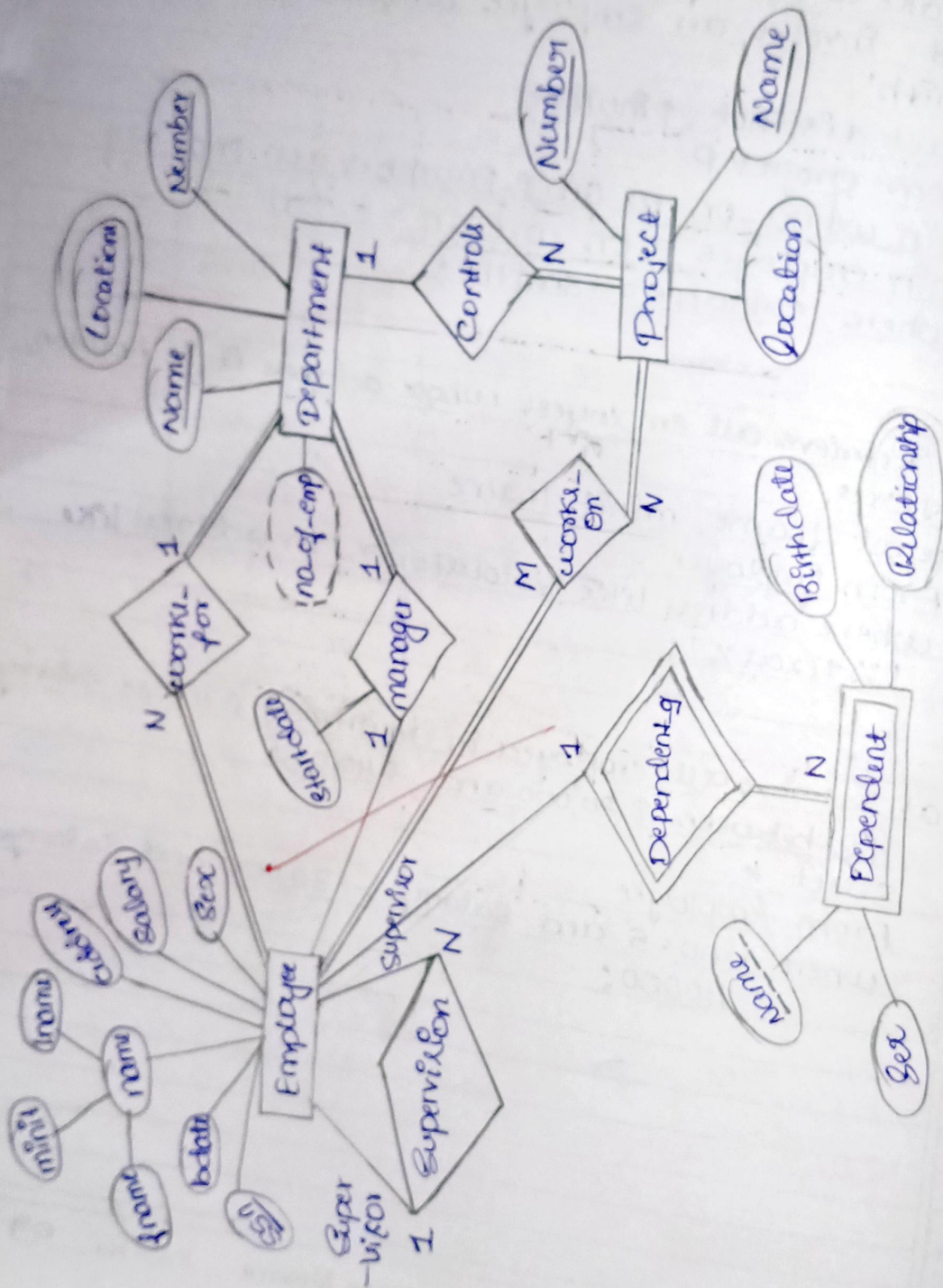
6. Retrieve ~~all~~ employees in department 5 whose salary is between \$20,000 and \$40,000.

Select *

From Employee

where dno = 5 and salary $\geq 20,000$ and salary ≤ 40000 ;

ER diagram for Company database





Result of mapping the Company ER schema into a relational schema.

Employee

fname	minf	lname	<u>SSN</u>	bdate	address	Sex	Salary	super SSN	dno
-------	------	-------	------------	-------	---------	-----	--------	--------------	-----

Department

dname	<u>dnumber</u>	mgrssn	mgrstartdate
-------	----------------	--------	--------------

Dept_locations

<u>Dnumber</u>	Dlocation
----------------	-----------

Project

Pname	<u>Pnumber</u>	Plocation	Dnum
-------	----------------	-----------	------

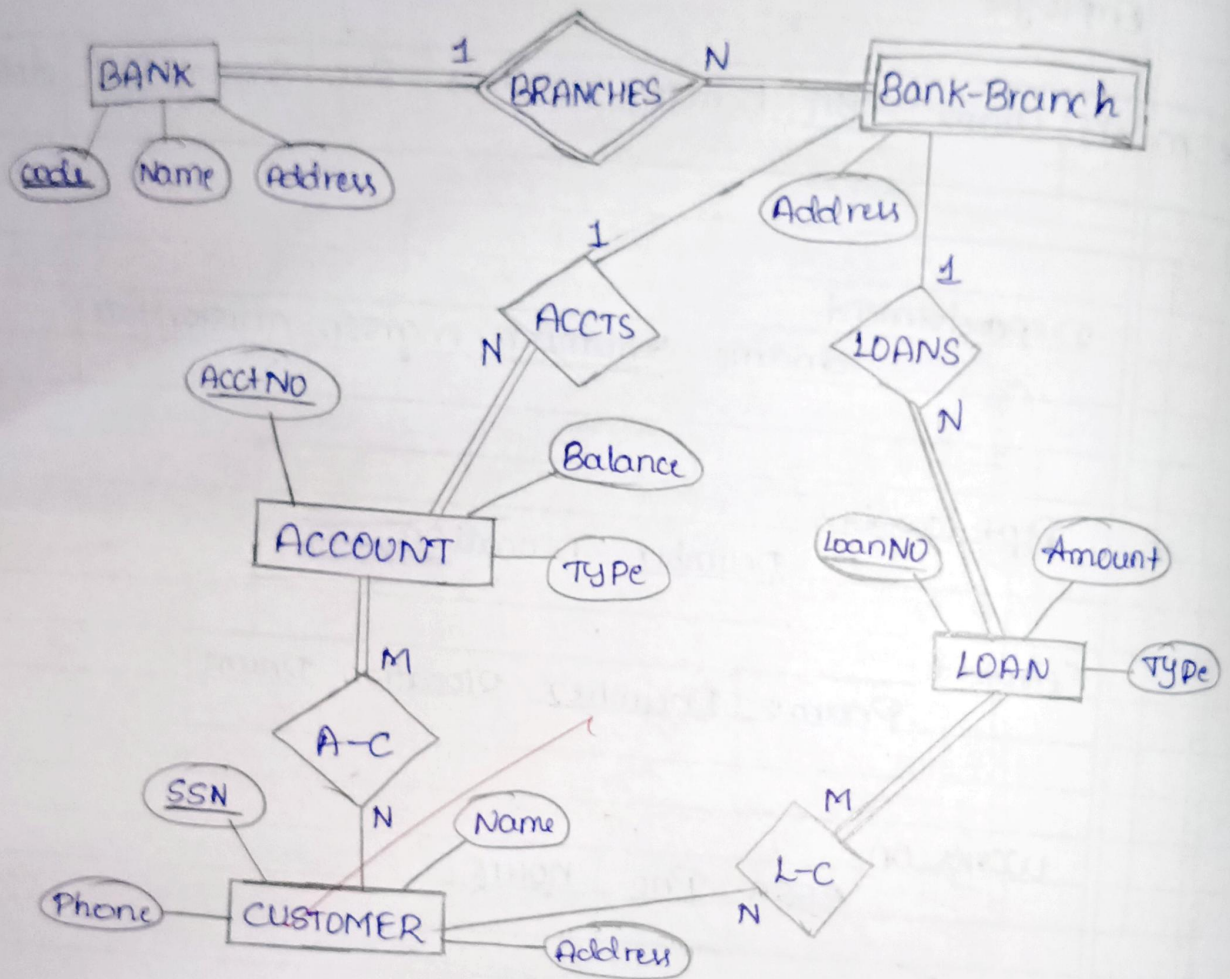
WORKS_ON

<u>ESSN</u>	<u>Pno</u>	hours
-------------	------------	-------

Dependent

<u>ESSN</u>	<u>Dependent_name</u>	Sex	bdate	relationship
-------------	-----------------------	-----	-------	--------------

ER diagram for Bank database





Date.....

Experiment No.....

Relational database schema of Bank database.

Bank

<u>code</u>	name	address
-------------	------	---------

Bank Branch

Address	BranchNO
---------	----------

Account

<u>AccountNo</u>	Balance	BranchNo.	Type.
------------------	---------	-----------	-------

Loan

<u>LoanNo</u>	Amount	Type	BranchNo.
---------------	--------	------	-----------

Customer

<u>Ssn</u>	Phone	AccNo	LoanNo	Name	address.
------------	-------	-------	--------	------	----------

✓

Desc Branch;

Field	Type	NULL	Key	Default	Extra
branch_name	varchar(25)	NO	PRI	NULL	
branch-city	varchar(20)	NO		NULL	
assetg	decimal(10,2)	NO		NULL	

Desc Account;

Field	Type	NULL	Key	Default	Extra
accno	int(11)	NO	PRI	NULL	
branch_name	varchar(25)	NO	MUL	NULL	
balance	decimal(10,2)	NO		NULL	

2.

1

Consider the following database for a banking enterprise.

BRANCH (branch_name: string, branch_city: string, assets: real)

ACCOUNT (accno: int, branch_name: string, balance: real)

CUSTOMER (customer_name: string, customer_street: string, customer_city: string)

DEPOSITOR (customer_name: string, accno: int)

LOAN (loan_number: int, branch_name: string, amount: real)

BORROWER (customer_name: string, loan_number: int)

Create all the above tables by properly specifying the primary keys & the foreign keys.

Create table Branch (

branch_name varchar(25) primary key,
branch_city varchar(20) not null,
assets decimal(10,2) not null
);

Create table Account (

accno int primary key,
branch_name varchar(25) not null,
balance decimal(10,2) not null,
foreign key (branch_name) references branch(branch_name)
);

Desc customer;

Field	Type	NULL	Key	default	Extra
customer_name	varchar(25)	NO	PRI	NULL	
customer_street	varchar(25)	NO		NULL	
customer_city	varchar(25)	NO		NULL	

Desc Depositor;

Field	Type	NULL	Key	Default	Extra
customer_name	varchar(25)	NO	MUL	NULL	
accno	Pnt(11)	NO	MUL	NULL	

Desc Loan;

Field	Type	NULL	Key	default	Extra
loan-number	Pnt(11)	NO	PRI	NULL	
branch-name	varchar(25)	NO	MUL	NULL	
amount	decimal(10,2)	NO		NULL	

Desc Borrower;

Field	Type	NULL	Key	Default	Extra
customer_name	varchar(25)	NO	PRI	NULL	
loan-number	Pnt(11)	NO	PRI	NULL	

create table customer(
customer-name varchar(25) primary key,
customer-street varchar(25) not null,
customer-city varchar(25) not null
);

create table Depositor(
customer-name varchar(25) not null,
accno.int not null,
foreign key (customer-name) references customer
(customer-name),
foreign key (accno) references account(accno) on
delete cascade
);

create table Loan(
loan-number int primary key,
branch-name varchar(25) not null,
amount decimal(10,2) not null,
foreign key (branch-name) references branch(branch-name)
);

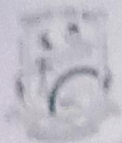
create table Borrower(
customer-name varchar(25) not null,
loan-number int not null,
foreign key (customer-name) references customer
(customer-name),
foreign key (loan-number) references loan(loan-number),
foreign key (customer-name, loan-number)
);

Select * from Branch;

Branch-name	Branch-city	assets
Basavanagudi	Bangalore	250000.00
Green park	delhi	300000.00
Jaynagar	bangalore	150000.00
Marin drive	Mumbai	400000.00
Noida	delhi	500000.00

Select * from Account;

accno	branch-name	balance
123	Jaynagar	25000.00
156	Jaynagar	30000.00
189	Basavanagudi	48888.00
436	Basavanagudi	15000.00
478	Marine drive	48000.00
778	Green park	60000.00
789	Noida	25000.00



2. Enter atleast five tuples for each relation.

Branch

Insert Into Branch values ('Jayanagar', 'bangalore',
15000000);

Insert Into Branch values ('Basavanagudi',
'bangalore', 25000000);

Insert Into Branch values ('Noida', 'delhi',
30000000);

Insert Into Branch values ('Marine drive', 'Mumbai',
40000000);

Insert Into Branch values ('Green park', 'delhi',
30000000);

Account

Insert Into Account values (123, 'Jayanagar', 25000)

Insert Into Account values (156, 'Jayanagar',
30000);

Insert Into Account values (456, 'Basavanagudi',
15000);

Insert Into Account values (789, 'Noida', 25000)

Insert Into Account values (478, 'Marine drive',
48000);

Insert Into Account values (778, 'Green park',
60000);

Insert Into Account values (189, 'Basavanagudi',
48888);

Select * from Customer;

customer_name	customer_street	customer_city
John	Noida	delhi
Kumar	Baravangudi	Bangalore
Mike	Marine drive	mumbai
Ramu	Jaynagar	bangalore
Sachin	Green Park	delhi

Select * from Depositor;

customer_name	aceno
Ramu	123
Ramu	156
Ramu	189
Kumar	456
John	789
Mike	478
Sachin	778

Customer

Insert into customer values ('Ramu', 'Jaynagar',
'bangalore');
Insert into customer values ('Kumar', 'Balavanagudi',
'bangalore');
Insert into customer values ('John', 'Noida', 'delhi');
Insert into customer values ('Mike', 'Marine drive',
'mumbai');
Insert into customer values ('Sachin', 'Green park',
'delhi');

Depositor

Insert into depositor values ('Ramu', 123);
Insert into depositor values ('Ramu', 156);
Insert into depositor values ('Ramu', 189);
Insert into depositor values ('Kumar', 456);
Insert into depositor values ('John', 789);
Insert into depositor values ('Mike', 478);
Insert into depositor values ('Sachin', 778);

Select * from loan;

loan-number	branch-name	amount
1111	Jaynagar	250000.00
2222	Balavanagudi	350000.00
3333	Noida	150000.00
4444	Marine drive	150000.00
5555	Greenspark	750000.00

Select * from Borrower.

customer-name	loan-number
Ramu	1111
Kumar	2222
John	3333
Mike	4444
Sachin	5555

Loan

Insert into loan values (1111, 'Jaynagar', 250000);
Insert into loan values (2222, 'Bachavanagudi',
350000);

Insert into loan values (3333, 'Noida', 150000);

Insert into loan values (4444, 'Marindrive',
1500000);

Insert into loan values (5555, 'Green park',
7500000);

Borrower

Insert into borrower values ('Ramu', 1111);

Insert into borrower values ('Kumar', 2222);

Insert into borrower values ('John', 3333);

Insert into borrower values ('Mike', 4444);

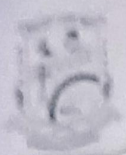
Insert into borrower values ('Sachin', 5555);

1)

customer_name	Count(*)
Ramu	2

2)

customer_name
Ramu

Queries:-

1. Find all the customers who have atleast two accounts to the main Branch.

```
Select distinct (customer_name), count(*)  
from account a, depositor d  
where a.accno = d.accno  
and d.accno in (select accno from account  
where branch name = "Jaynagar")  
group by d.customer_name  
having count(*) >= 2;
```

2. Find all the Customers who have an account at all the branches located in a Specified city.

```
Select d.customer_name  
from account a, depositor d, branch b  
where b.branch_name = a.branch_name and  
a.accno = d.accno  
and b.branch_city = "Bangalore"  
group by customer_name;  
having count (distinct b.branch_name) = (Select  
count (branch_name)  
from branch  
where branch_city = "Bangalore");
```


8) 1 row affected.

Select * from account;

Aceno	branch_name	balance
123	Tayragan	25000.00
156	Tayragan	30000.00
189	Basavaraqudi	48888.00
456	Basavaraqudi	15000.00
478	Marina drive	48000.00

Select * from depositor.

customer_name	Aceno
Ramu	123
Ramu	156
☒ Ramu	189
Kumar	456
Mike	478



Date.....

Experiment No.....

3. Demonstrate how you delete all account tuples at every branch located in a specific city.

```
delete from account  
where branch_name in (select branch_name)  
from branch  
where branch_city = "delhi" ;
```

~~1~~

14

fname	Lname
Ritvik	Rajput

31

fname	Lname
Ankur	Singh

21

fname	Lname
Mariya	Moya
Ritvik	Rajput
Shivanya	Sharma



Complex Queries.

- i) Retrieve the names of all employees who do not have Supervisors.

```
Select e.fname, e.lname  
from employee as e  
where NOT exists (select *  
from employee as s  
where e.superssn = s.ssn);
```

- ii) Retrieve the name of each employee who has a dependent with the same first name and is the same gender as employee

```
Select e.fname, e.lname  
from employee as e (select *  
from dependent d  
where e.fname = d.dependent name and  
e.sex = d.sex and e.ssn = d.ssn);
```

- iii) Retrieve the names of employees who have no dependents.

```
Select e.fname, e.lname  
from employee as e  
where NOT exists (select *  
from dependent as d  
where e.ssn = d.ssn);
```


fname	Lname
John	Smith
Mariya	Moya
Ritvik	Rajput
Ghivanya	Sharma

5)

SSN
12345678
12345555
45678821

6)

total-salary	max-salary	min-salary	avg-salary
130000.00	50000.00	30000.00	43333.33



- iv) List the names of managers who have atleast one dependent.

```
select b.fname, b.lname  
from employee b  
where (select count(*)  
from dependent where b.ssn = e.ssn) >= 1;
```

- v) Retrieve the Social Security numbers of all employees who work on project numbers 1, 2 or 3.

```
select e.ssn  
from employee e  
where e.ssn in (select essn  
from works_on  
where pnumber = 1 or pnumber = 2 or pnumber = 3);
```

- vi) Find the sum of the salaries of all employees of the 'Research' department as well as the maximum salary, the minimum salary, and the average salary in this department.

```
select sum(salary), max(salary), min(salary),  
avg(salary)  
from employee, department  
where dname = 'Research' and dno = department;
```


#1

dno	num-employees	avg-salary
5	1	30000.00
6	1	40000.00
7	1	50000.00
8	1	60000.00
9	1	70000.00



Date.....

Experiment No.....

vii)

For each department, retrieve the department number, the number of employees in the department, and their average salary.

Select dno, count(*), avg(salary)
from employee
group by dno;

~~15/11/24~~